

Jordan 2

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Problem 1. Now we assume A has all its eigenvalues in F and show that it is conjugate to a Jordan form (which you should look up). Explain why we can assume A is nilpotent, and that in this case, the existence of Jordan form is equivalent to showing we have a basis of form

$$w_1, Aw_1, \dots, A^{s_1}w_1,$$

$$w_2, Aw_2, \dots, A^{s_2}w_2$$

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$$w_k, Aw_k, \dots, A^{s_k}w_k, \text{ with } A^{s_i}w_i \text{ in Ker } A.$$

To prove it, use induction on the dimension. First consider $A(V) \subset V$. Explain why this is lower dimensional and $A : A(V) \rightarrow A(V)$ is nilpotent, so induction applies. Now use induction.

Solution. Let A be any matrix with all its eigenvalues in F . By the previous homework $\oplus_{f \in F} GV_f(A) = V$ thus we know that $(A - fI)$ for f being a generalized eigenvector is nilpotent. If $(A - fI)$ has a Jordan form then A has a Jordan form. Thus assume A is nilpotent.

Since A is nilpotent let k be the smallest natural number such that $A^k = 0$

$A(V) \subset V$ where $\dim(V) > \dim A(V)$, because $\forall v \in V \ A^{k-1}v \in \ker A$. Thus $\dim \ker A \neq 0$. Thus $A(V)$ is lower dimensional than V .

$\forall v \in A(V), Av \in A(V)$. Thus $A : A(V) \rightarrow A(V)$.

Base Case: Suppose $\dim V = 1$, then $\forall v \in V$ s.t $v \neq 0$. Thus $Av = 0$. Thus v a basis of the form seen above.

Inductive Hypothesis: Assume all vector spaces V with $\dim V \leq k$ for any nilpotent matrix A have a basis of the form above

Inductive Step: Let V be a vector space with $\dim V = k + 1$. $\dim A(V) \leq k$. Thus it has a basis of the form above let this basis be B . For all $w_i \in B$, define $z_i = Aw_i$ where $z_i \in V$. Thus for all $\{z_i, Az_i (= w_i), \dots, A^{s_i}z_i | A^{s_i}z_i \neq 0, A^{s_i+1}z_i = 0, i \in \mathbb{N}\} = B$. Create a linear combination of B equal to 0 with nonzero coefficients. Apply A to the linear combination. All $A^{s_i}z_i$ will be killed and all A^jz_i for $j < s_i$ will become $A^{j+2}z_i$. This new linear combination will contain all the elements of B except z_i and will only equal 0 if the coefficients are zero thus the elements are linearly independent. $A^{s_i}z_i$ are linearly independent because they are parts of basis. Thus B is linearly independent. $\forall x \in V, Ax = As$ s.t $s \in \text{span}(B)$ thus $A(x - s) = 0$ and $x - s \in \ker A$. Thus there are elements which are not in the span of B . $x \in \ker A - \text{span} B$ x is linearly independent from B . Furthermore since $x \in \ker A$ it satisfied our condition such that $Ax = 0, B = B \cup x$. Then check again if there exists $x \in \ker A - \text{span} B$ and if it does add it into B . Thus $\dim \text{span} B = \dim \ker A \dim A(V) = V$ and is linearly independent and thus is a basis of the required form.

Thus by induction any vectorspace has a basis of the required form and thus is conjugate to a Jordan form. ■

Problem 2. Assuming A has all its eigenvalues in F , there exist N, S , N nilpotent, S diagonalizable with the properties

a) $N + S = A$

b) N and S commute with every transformation that commutes with A (in particular they commute with A and with each other) Moreover, N and S are unique.

Show first that N and S exist (eg use the Jordan form)

Proof. Since A has eigenvalues in F . A has a Jordan form $A = PJP^{-1}$ where J is its Jordan form. Write $J = D + N$ where D is its diagonal entries and N is the rest. N is strictly upper triangular since J is upper triangular. Thus N is nilpotent. Thus there exists a k such that $N^k = 0$. $A = PJP^{-1} = P(D + N)P^{-1}$ and since conjugation is automorphism $A = PDP^{-1} + PNP^{-1}$. $S = PDP^{-1}$ is a diagonalizable matrix by definition. Furthermore, $(PNP^{-1})^k = \prod_{i=1}^k PNP^{-1} = P \prod_{i=1}^{k-1} N(P^{-1}P)NP^{-1} = PN^kP^{-1} = 0$. Thus PNP^{-1} is nilpotent. $\square$

Problem 3. Suppose A and B commute, show that they preserve each others eigenspaces, and if A is diagonalizable the converse holds (if B preserves A 's eigenspaces, then B commutes with A).

Proof. ($\implies$): Suppose A and B commute. Let f be an eigenvalue of A . $\forall v \in V_f(A)$ thus $Av = fv$. Then $BAv = Bfv = fBv$ but since $AB = BA$, $BAv = ABv = fBv$ thus $Bv \in V_f(A)$. Let g be an eigenvalue of B . $\forall v \in V_g(B)$ thus $Bv = gv$. Then $ABv = Agv = gAv = BAv$ thus $Av \in V_g(B)$. Thus A and B preserve each others eigenspaces.

($\impliedby$): Suppose A is diagonalizable and that B preserves A 's eigenspaces. Then thus $\forall v \in V_f(A)$ $Bv \in V_f(A)$. Since A is diagonalizable there exists a basis of eigenvectors $\{v_1, \dots, v_{\dim V}\}$ where the eigenvalue for v_i is f_i . Then for all $v \in V$, $v = \sum_{i=1}^{\dim V} a_i v_i$ and $Bv = \sum_{i=1}^{\dim V} a_i Bv_i$ and $ABv = \sum_{i=1}^{\dim V} a_i ABv_i$ but since B preserves the eigenspace, $ABv = \sum_{i=1}^{\dim V} a_i f_i Bv_i = \sum_{i=1}^{\dim V} a_i Bf_i v_i = B \sum_{i=1}^{\dim V} a_i f_i v_i = B \sum_{i=1}^{\dim V} a_i Av_i = BA \sum_{i=1}^{\dim V} a_i v_i = BAv$. Thus $\forall v \in V$, $ABv = BAv$ thus $AB = BA$. $\square$

Problem 4. Suppose A is diagonalizable and W is a subspace of V preserved by A . Explain why $A|_W : W \rightarrow W$ is diagonalizable.

Proof. Since A is diagonalizable there is an eigenvector basis $B = \{v_1, \dots, v_{\dim V}\}$. Form a subspace W' with dimension $\dim W$ by taking the basis $B' = \{v_1, \dots, v_{\dim W}\}$, there are $\dim W$ linearly independent vectors. Let the basis of W be $\{w_1, \dots, w_{\dim W}\}$. Create a linear map $Q : W' \rightarrow W$ thus $v_i \rightarrow w_i$ for $i \in [1, \dots, \dim W]$ Q is a bijection because it maps distinct bases in W' to distinct bases in W . Since W' has an eigenvector basis with regards to A then A restricted to W' is diagonalizable. $A = PDP^{-1}$. Furthermore A restricted to W , $A = QPDP^{-1}Q^{-1}$ let $R = QP$, thus A restricted to W is $A = RDR^{-1}$ and A is diagonalizable. $\square$

Problem 5. Suppose $A = N_1 + S_1 = N_2 + S_2$ with the above properties. So we have $N_1 - N_2 = S_2 - S_1$. Explain why $N_1 - N_2$ is nilpotent and $S_2 - S_1$ is diagonalizable. conclude $N_1 = N_2$ and $S_2 = S_1$.

Proof. Let $N_1^r = 0$ and $N_2^s = 0$. Since N_1 and N_2 commute with A and thus by the properties they must commute with each other. Thus the binomial theorem applies. $(N_1 - N_2)^{r+s} = \sum_{i=0}^{s+r} (-1)^i \binom{s+r}{i} N_1^{s+r-i} N_2^i$ thus $s+r-i > r$ or $i > s$. Thus $(N_1 - N_2)^{r+s} = 0$. Thus $N_1 - N_2$ is nilpotent. By the properties of S_1 and S_2 they commute with A and thus commute with each other. Thus they preserve each others eigenspaces by problem 3.

The only nilpotent diagonalizable matrix is 0. Thus $N_1 - N_2 = 0$ and $S_1 - S_2 = 0$. Thus $N_1 = N_2$ and $S_1 = S_2$. $\square$